

DAY 8

```
C:\Users\Adminuser>java -version
```

```
java version "21.0.11" 2026-04-21 LTS
```

```
Java(TM) SE Runtime Environment (build 21.0.11+9-LTS-211)
```

```
Java HotSpot(TM) 64-Bit Server VM (build 21.0.11+9-LTS-211, mixed mode, sharing)
```

```
C:\Users\Adminuser>mvn -v
```

```
Apache Maven 3.9.16 (2bdd9fddda4b155ebf8000e807eb73fd829a51d5)
```

```
Maven home: C:\apache-maven-3.9.16
```

```
Java version: 21.0.11, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-21.0.11
```

```
Default locale: en_US, platform encoding: UTF-8
```

```
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
```

```
-- cd C:\Users\Adminuser\Desktop\IBM
```

```
--->mvn archetype:generate -DgroupId=com.bank -
```

```
DartifactId=Demo02_DependencyInjection -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

```
-----cd Demo02_DependencyInjection
```

```
-----mvn clean install
```

```
-----mvn exec:java -Dexec.mainClass="com.bank.App"
```

Task 1:

What do you understand by Spring framework?

- Spring Framework is an open-source Java framework used for developing scalable enterprise applications. It simplifies application development through features like Dependency Injection (DI) and Inversion of Control (IoC). Spring provides comprehensive support for web development, database connectivity, transaction management, and security. Its modular architecture allows developers to use only the components they need. Overall, Spring helps build maintainable, flexible, and high-performance Java applications.

Task 2:

What is spring boot?

- Spring Boot is an extension of the Spring Framework that simplifies the development of Java applications by reducing configuration and setup time.
- It comes with built-in features like autoconfiguration. Spring Boot helps developers create standalone, ready-to-run applications quickly.
- It follows the convention-over-configuration approach, minimizing the need for XML configurations. Overall, Spring Boot makes Spring application development faster, easier, and more efficient.

Task 3:

What is the diff b/w spring framework and spring boot?

- Spring Framework provides the core features for Java application development, while Spring Boot builds on top of Spring Framework and makes it easier and faster to create and run Spring applications with minimal configuration.

Task 4:

What is Dependency Injection?

- Dependency Injection (DI) in Java is a design pattern where the objects required by a class are provided from outside rather than being created within the class itself. This helps reduce dependency between classes and promotes loose coupling.
- In Spring, the IoC container automatically creates and injects the required objects into a class.
- It allows developers to change or replace dependencies easily without modifying the core business logic.
- It promotes loose coupling between the classes, Makes code easier to test and maintain.

Task 5:

create a simple OrderService that depends on a PaymentService. Instead of creating objects manually, inject them
interfaces:

paymentService ' ----> credit card or debit card class
and orderservice ---> PaymentService implementation class

App.java

```

package com.bank;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class App {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext(
                "applicationContext.xml"
            );

        OrderService orderService =
            context.getBean(
                "orderService",
                OrderService.class
            );

        orderService.placeOrder(500);

    }
}

```

Paymentservice.java

```

package com.bank;

public interface PaymentService {

    void pay(double amount);

}

```

CreditCardPayment.java

```

package com.bank;

public class CreditCardPayment implements PaymentService {

    @Override
    public void pay(double amount) {

        System.out.println(
            "Payment of $" + amount + " done using Credit Card"
        );

    }

}

```

DebitCardPayment.java

```
package com.bank;

public class DebitCardPayment implements PaymentService {

    @Override
    public void pay(double amount) {

        System.out.println(
            "Payment of $" + amount + " done using Debit Card"
        );
    }
}
```

OrderService.java

```
package com.bank;

public class OrderService {

    private PaymentService paymentService;

    // Constructor Injection
    public OrderService(PaymentService paymentService) {

        this.paymentService = paymentService;
    }

    public void placeOrder(double amount) {

        System.out.println("Order placed");

        paymentService.pay(amount);
    }
}
```

applicationContext.xml

```
<bean id="paymentService"
    class="com.bank.CreditCardPayment"/>

<bean id="orderService"
    class="com.bank.OrderService">
```

OUTPUT

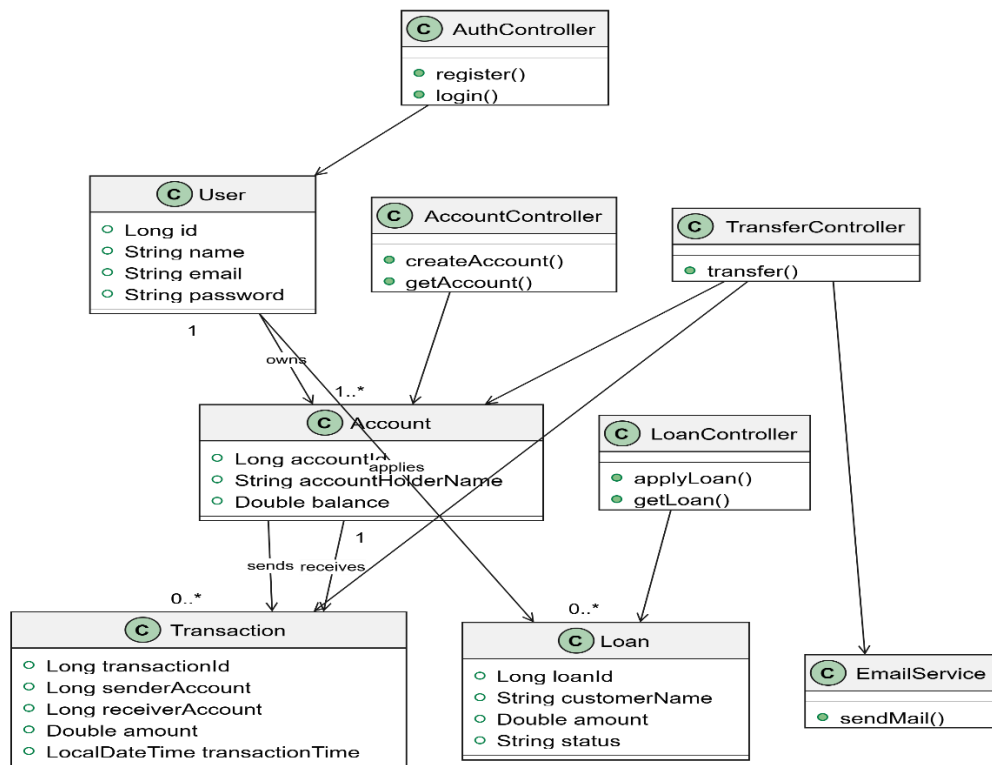
```
Administrator: Command Pro x + v
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ Demo02_DependencyInjection ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 10 source files with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.260 s
[INFO] Finished at: 2026-07-27T08:51:12Z
[INFO] -----

C:\Users\Adminuser\Desktop\IBM\Demo02_DependencyInjection>mvn exec:java -Dexec.mainClass="com.bank.App"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.bank:Demo02_DependencyInjection >-----
[INFO] Building Demo02_DependencyInjection 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec:3.6.3:java (default-cli) @ Demo02_DependencyInjection ---
Order placed
Payment of $500.0 done using Debit Card
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.446 s
[INFO] Finished at: 2026-07-27T08:51:21Z
[INFO] -----

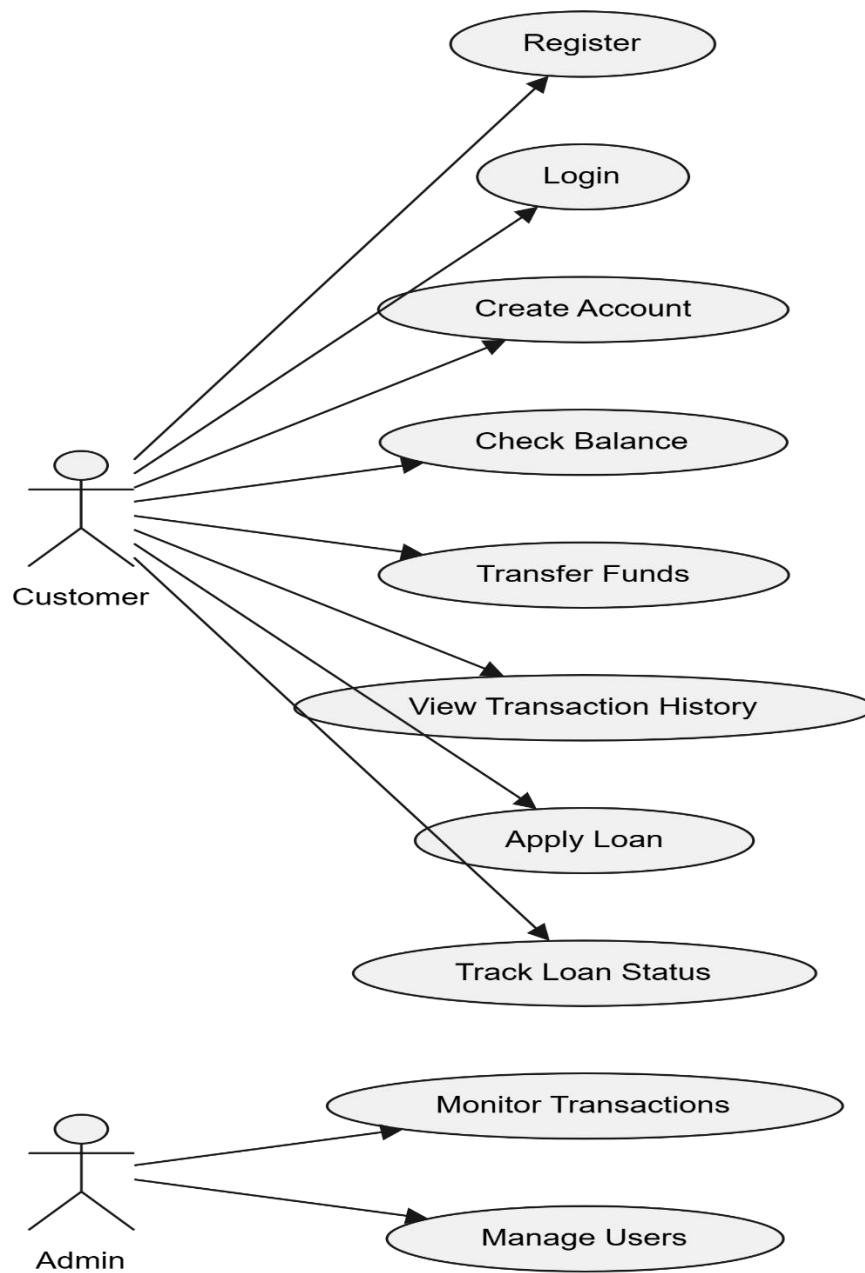
C:\Users\Adminuser\Desktop\IBM\Demo02_DependencyInjection>
```

UML DIAGRAM FOR BANK APPLICATION

- CLASS



Use Case Diagram



Sequence Diagram for Fund Transfer

